CLUSTERING

$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$

→ binary classification: $y_i \in \{0, 1\}$ or $y_i \in \{-1, +1\}$

→ multiclass classification: $y_i \in \{1, 2, \dots, K\}$

$\mathcal{X} = \{x_i\}_{i=1}^{N}$ → clustering NO CLASS LABELS !!!

PARAMETRIC CLASSIFICATION

- we assumed that each class follows a certain density

$p(x|y=c)$ ⇒ class conditional density

- we estimated the parameters

$p(x|y=1), p(x|y=2), \dots, p(x|y=K), Pr(y=1), Pr(y=2), \dots, Pr(y=K)$

$\hat{\mu}_1, \hat{\Sigma}_1$ $\quad \hat{\mu}_2, \hat{\Sigma}_2$ $\quad \hat{\mu}_K, \hat{\Sigma}_K$ $\quad \hat{Pr}(y=1)$ $\quad \hat{Pr}(y=2)$ $\quad \hat{Pr}(y=K)$

$\hat{Pr}(y=c|x)$ ⇒ predicted posterior probability

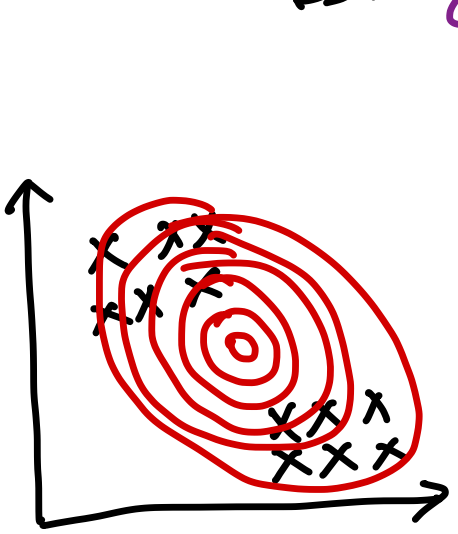
Mixture Densities

C_k = cluster k

$$p(x) = \sum_{k=1}^{K} p(x \mid C_k) Pr(C_k)$$

mixture proportions

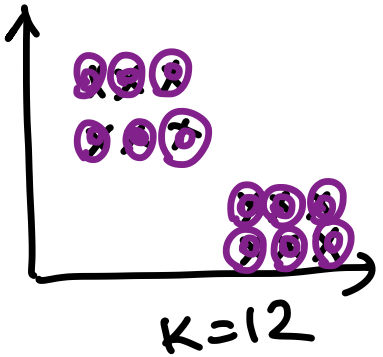
component densities



K = 1

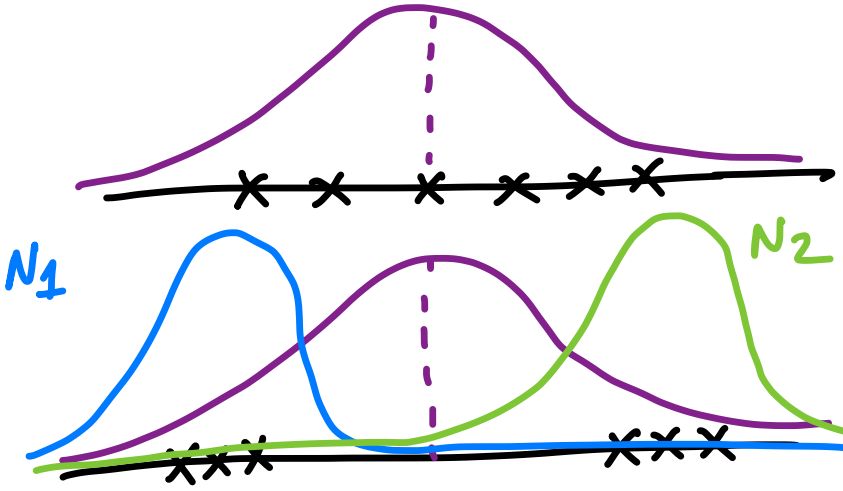


K = 2



K = 12

K = # of clusters / components / groups / segments / ⋮



$0.5 N_1 + 0.5 N_2$

≥ 0 ≥ 0

mixture density

$$\int_{-\infty}^{+\infty} [0.5 p_1(x) + 0.5 p_2(x)] dx$$

$$0.5 \underbrace{\int_{-\infty}^{+\infty} p_1(x) dx}_{1} + 0.5 \underbrace{\int_{-\infty}^{+\infty} p_2(x) dx}_{1}$$

$$\underbrace{\pi}_{\geq 0} \cdot N_1 + \underbrace{(1-\pi)}_{\geq 0} N_2$$

$$y_{ik} = \begin{cases} 1 & \text{if } y_i \text{ belongs to component/cluster/group/segment } k \\ 0 & \text{otherwise} \end{cases}$$

$$\Phi = \left\{ \hat{Pr}(C_k), \hat{\mu}_k, \hat{\Sigma}_k \right\}_{k=1}^{K}$$

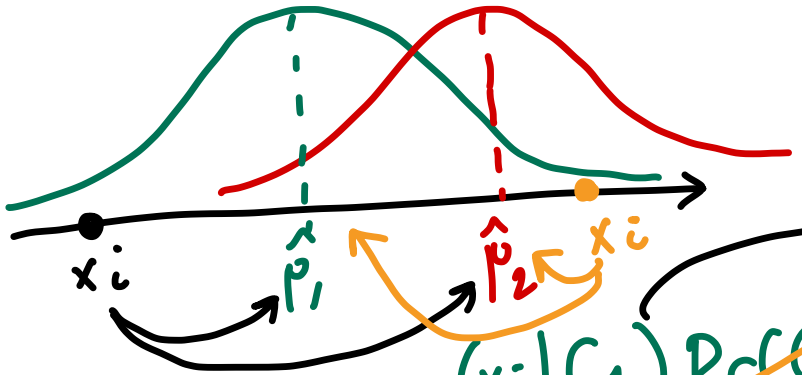
Iterative Algorithm

STEP#1: Estimate the cluster memberships $(\hat{y}_{ik})$

STEP#2: Estimate the other parameters (Φ)

$$\hat{Pr}(C_k) = \frac{\sum_{i=1}^{N} \hat{y}_{ik}}{N} \qquad \hat{\mu}_k = \frac{\sum_{i=1}^{N} \hat{y}_{ik} \cdot x_i}{\sum_{i=1}^{N} \hat{y}_{ik}} \qquad \hat{\Sigma}_k = \frac{\sum_{i=1}^{N} \hat{y}_{ik} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T}{\sum_{i=1}^{N} \hat{y}_{ik}}$$

K-MEANS CLUSTERING



Assuming (i) $Pr(C_1) = Pr(C_2)$

(ii) $\sigma_1^2 = \sigma_2^2$

$$Pr(C_1|x_i) = \frac{p(x_i|C_1)\,Pr(C_1)}{p(x_i)} \qquad \frac{1}{\sqrt{2\pi\hat{\sigma}_1^2}} \cdot \exp\left[-\frac{(x_i - \hat{\mu}_1)^2}{2\hat{\sigma}_1^2}\right]$$

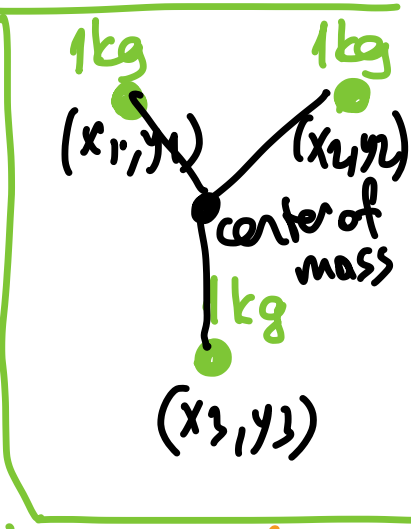
$$Pr(C_2|x_i) = \frac{p(x_i|C_2)\,Pr(C_2)}{p(x_i)} \qquad \frac{1}{\sqrt{2\pi\hat{\sigma}_2^2}} \cdot \exp\left[-\frac{(x_i - \hat{\mu}_2)^2}{2\hat{\sigma}_2^2}\right]$$

Compare $(x_i - \hat{\mu}_1)^2$ & $(x_i - \hat{\mu}_2)^2$

$\|x_i - \hat{\mu}_1\|_2^2$ & $\|x_i - \hat{\mu}_2\|_2^2$ } Squared Euclidean distance

$$\text{Error} = \sum_{i=1}^{N}\sum_{k=1}^{K} b_{ik}\|x_i - \hat{\mu}_k\|_2^2$$ ⇒ total squared distance to nearest centroids

$$b_{ik} = \begin{cases} 1 & \text{if } \|x_i - \hat{\mu}_k\|_2 = \min_{c=1}^{K} \|x_i - \hat{\mu}_c\|_2 \\ 0 & \text{otherwise} \end{cases}$$

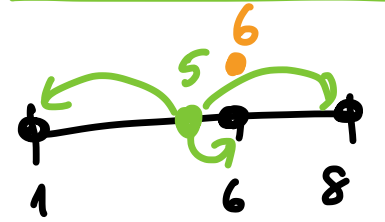


minimize $\sum_{i=1}^{N}\sum_{k=1}^{K} b_{ik}\|x_i - \hat{\mu}_k\|_2^2$

with respect to: $\hat{\mu}_1, \hat{\mu}_2, \dots, \hat{\mu}_k, \{b_{ik}\}_{i=1,k=1}^{N,K}$

- Initialize $\hat{\mu}_1, \hat{\mu}_2, \dots, \hat{\mu}_k$ randomly
- Repeat [
  - for all x_i: $b_{ik} = \begin{cases} 1 & \text{if } \|x_i - \hat{\mu}_k\|_2 = \min_{c=1}^{K} \|x_i - \hat{\mu}_c\| \\ 0 & \text{otherwise} \end{cases}$ (assign each x_i to its nearest center)
  - for all $\hat{\mu}_k$: $\hat{\mu}_k = \dfrac{\sum_{i=1}^{N} b_{ik} \cdot x_i}{\sum_{i=1}^{N} b_{ik}}$

- Until convergence [if all b_{ik}'s stay the same] or [if all $\hat{\mu}_k$'s stay the same]



$(6-5)^2 + (8-5)^2$

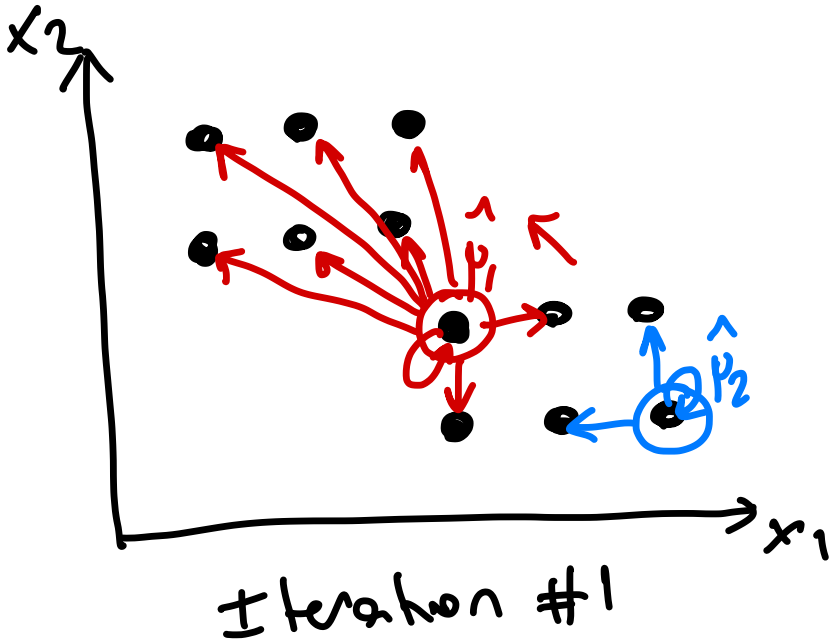
$(1-5)^2 = 26$

$(6-6)^2 + (6-1)^2$

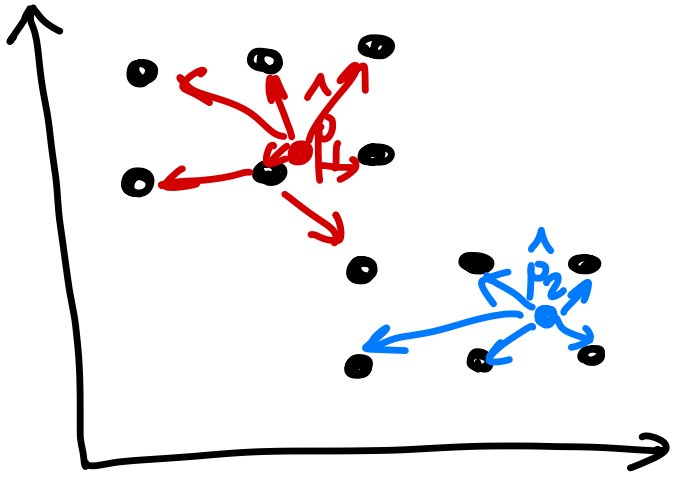
$(8-6)^2 = 29$

$\min (x_i - c)^2$ → sample mean

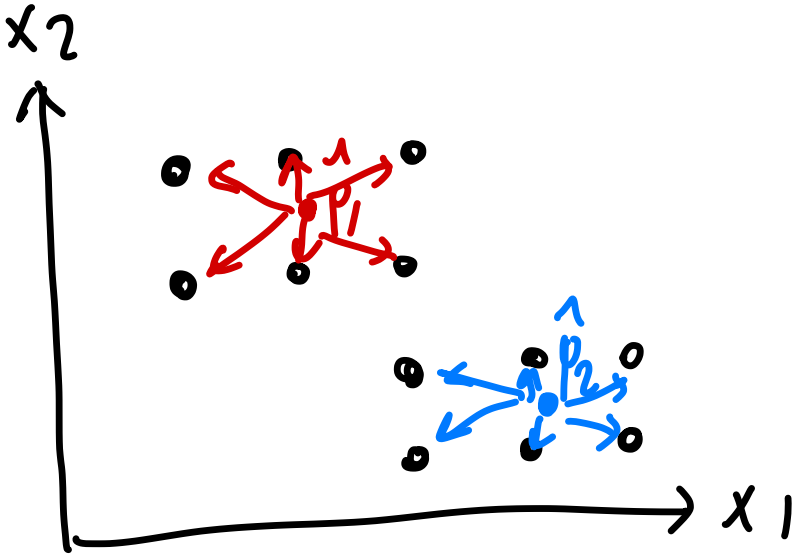
minimize $|x_i - c|$ → sample median



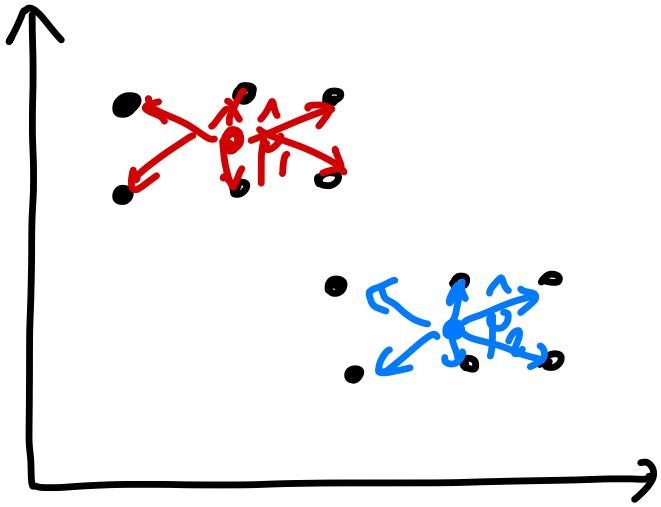
Iteration #1



Iteration #2



Iteration #3



Iteration #4